

ABU NAKHLAH, Qatar

WMO: 411705

Lat: 25.117N

Lon: 51.315E

Elev: 130

StdP: 14.63

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.9	51.5	24.0	18.2	91.0	28.0	22.1	91.0	27.3	64.5	25.4	64.8	7.9	280	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.6	115.2	70.4	113.4	70.7	111.8	70.9	85.6	94.2	84.6	94.1	83.6	93.9	13.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.9	178.3	91.6	82.4	169.5	91.8	80.8	160.7	91.5	50.5	93.6	49.3	94.2	48.0	93.8	89.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.3	24.7	21.6	44.1	118.7	3.2	1.2	41.8	119.6	39.9	120.3	38.1	121.0	35.8	121.9
			39.6	87.6	3.6	1.3	37.0	88.6	34.9	89.3	32.8	90.1	30.2	91.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	63.2	66.8	73.8	83.0	92.6	96.9	99.2	98.9	93.9	86.3	76.1	66.7
	DBStd	13.55	4.78	5.62	5.29	5.31	3.92	3.01	2.36	2.11	3.01	3.80	4.88	4.75
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	176	94	41	3	0	0	0	0	0	0	0	1	37
	CDD50	12117	408	469	736	990	1321	1406	1526	1517	1317	1126	783	518
	CDD65	6817	37	90	274	540	856	956	1061	1052	867	661	333	90
	CDH74	102107	215	630	2499	6566	13605	16235	18389	17789	13795	8903	2982	499
	CDH80	69572	28	179	1072	3678	9364	11944	13926	13325	9554	5236	1195	71
Wind	WSAvg	9.5	9.1	10.2	10.7	10.1	10.1	11.8	10.6	8.3	8.1	7.3	8.3	9.3
Precipitation	PrecAvg	3.2	0.6	0.4	0.7	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.7
	PrecMax	9.9	2.1	2.7	5.5	0.6	0.4	0.0	0.0	0.0	0.0	0.3	4.3	3.5
	PrecMin	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.6	0.6	1.2	0.2	0.1	0.0	0.0	0.0	0.0	0.1	1.1	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.0	91.3	98.8	106.0	114.6	117.0	118.1	116.5	112.7	105.7	96.6	85.3
		MCWB	61.0	63.3	63.3	66.3	68.2	69.9	71.5	71.0	70.1	69.2	66.6	63.0
	2%	DB	79.0	85.4	93.5	102.8	111.5	114.9	115.2	114.5	109.9	102.5	92.8	82.0
		MCWB	61.2	62.7	62.6	66.2	68.3	69.1	71.0	71.8	71.6	70.4	67.1	64.1
	5%	DB	76.1	80.8	89.8	100.0	109.2	112.9	113.3	112.7	108.0	100.5	89.8	79.1
		MCWB	60.4	61.3	62.3	65.9	68.2	68.6	71.4	72.5	72.0	69.7	67.6	63.6
	10%	DB	73.2	77.3	85.9	96.4	106.1	110.9	111.4	110.8	106.0	98.2	87.1	76.8
		MCWB	59.4	60.1	62.1	65.2	67.8	68.9	71.5	73.3	72.3	70.8	67.4	63.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.4	69.7	70.9	73.6	78.6	82.3	86.2	87.5	85.4	81.2	75.8	71.5
		MCDB	72.0	74.8	77.6	85.0	87.9	92.8	94.6	94.3	94.2	90.3	83.5	75.9
	2%	WB	65.6	67.4	69.3	71.9	76.2	79.7	84.8	86.3	83.9	79.6	74.0	69.2
		MCDB	71.7	73.2	78.0	84.3	89.5	91.2	94.5	94.3	92.8	89.7	82.3	74.3
	5%	WB	63.9	65.7	67.6	70.4	74.5	77.6	83.6	85.3	82.7	78.3	72.3	67.4
		MCDB	70.1	73.3	77.5	84.4	90.7	92.8	94.7	94.2	92.0	88.4	81.0	73.6
	10%	WB	62.4	64.1	66.1	69.2	72.8	75.5	82.1	84.4	81.5	77.1	70.8	65.6
		MCDB	69.1	73.1	77.3	84.4	91.2	93.1	94.9	94.0	91.7	87.0	80.0	72.8
Mean Daily Temperature Range	5% DB	MDBR	18.4	19.7	22.5	23.4	26.7	27.6	25.6	24.9	25.4	23.3	19.0	17.6
		MCDBR	21.7	23.7	28.2	28.1	30.1	30.4	28.2	27.6	27.5	25.7	22.5	20.3
		MCWBR	8.9	8.3	7.8	7.8	8.9	10.0	11.5	12.7	12.2	11.1	9.2	8.5
	5% WB	MCDBR	18.4	19.5	21.1	22.8	27.0	28.3	24.5	23.4	24.5	23.0	20.9	18.0
		MCWBR	9.3	8.4	8.0	8.2	10.2	11.8	11.4	11.7	11.8	10.5	9.5	8.7
Clear-Sky Solar Irradiance	taub	0.509	0.557	0.624	0.671	0.665	0.642	0.690	0.647	0.569	0.511	0.507	0.486	
	taud	1.823	1.716	1.582	1.534	1.530	1.562	1.533	1.617	1.742	1.858	1.870	1.890	
	Ebn at Noon	229	229	222	216	217	221	210	219	233	239	228	230	
	Edh at Noon	60	71	85	91	91	88	90	83	72	61	56	54	
All-Sky Solar Radiation	RadAvg	1212	1462	1690	1875	2124	2238	2116	2041	1905	1630	1275	1140	
	RadStd	73	75	102	129	58	63	57	45	36	39	58	71	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.99	N/A	-2.92	+1.26	N/A	N/A	N/A	N/A	+378	+365

Nomenclature: See separate page